

Thapar Institute of Engineering and Technology, Patiala

Computer Science and Engineering Department

Course Name: Object Oriented Programming

Course Code:UTA028

Time: 120 Minutes

Maximum Marks: 30

Coordinator: Karun Verma

Note: 1. Attempt all the questions

2. Start answering each question on a separate page

No.	Question	Marks	CO	BL
Q1.	Describe the following by citing examples. a. Encapsulation b. Inheritance c. Abstraction d. Polymorphism	1.5*4	CO3	L2
Q2.	a. How are arrays declared in Java? b. “ Counting sort is a linear time sorting algorithm that counts the occurrences of each element in an array and uses this information to determine the sorted order. The intuition behind using counting sort to find the K^{th} smallest element is to take advantage of its counting phase, which essentially calculates the cumulative frequencies of elements. By tracking these cumulative frequencies and finding the point where the count reaches or exceeds K can determine the K^{th} smallest element efficiently.”. The algorithm is: - Find the maximum element in the array to know the range. - Declare an array freq to store frequency of each element in the input array. - Iterate through the input array and update the frequencies of elements in the freq array. - Initialize a variable count to keep track of cumulative frequencies of elements. - Iterate through the freq array, if the frequency of current element is non-zero, increment the counter. - Return the current element if count is greater than or equal to K. Write a program in Java to implement the Counting sort algorithm above and return the K^{th} smallest element of the array.	1+5	CO1 CO2	L2 L4
Q3.	Write a program in Java having a class Tracker that tracks the footsteps of a user. Tracker should have three private data members: <i>PersonId</i> , <i>step_count</i> , <i>stair_count</i> . The class should satisfy the following requirements: a. A parameterized constructor should be defined to initialize data members. <i>PersonId</i> should be input by the user, whereas <i>step_count</i> and <i>stair_count</i> if not known may be set as 0 using constructors with default arguments.	2*3	CO2	L4

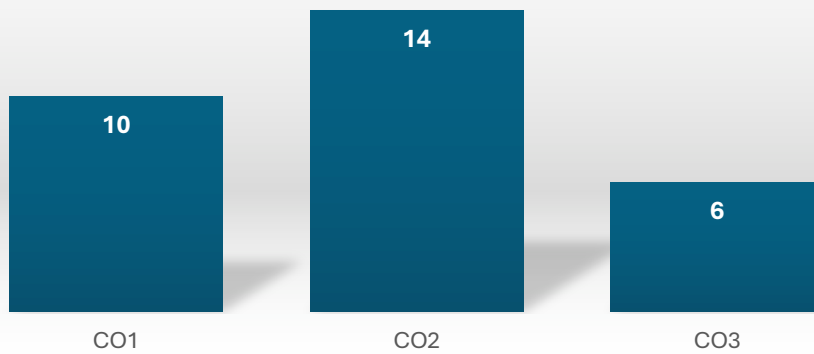
	b. Write member method <i>addSteps(int n)</i> and <i>addStairs(int n)</i> for incrementing the two counts. The <i>addStairs</i> method has the following assumption ($1 \text{ stair_count} = 2 \text{ step_count}$) c. In addition, the class must use array of objects for creating the objects and track how many Tracker objects are created. A means for querying this count should be provided. The code must not use any global variables. (Hint: Use static members and static functions to display the count of Tracker objects.)			
Q4.	a. What are different types of variables in Java? Explain the properties of each of them with suitable examples. b. What is the difference between a primitive variable and a reference variable? What is the size of a reference variable? What are the different kinds of operations allowed on reference variables?	3	CO1	L2
		3	CO1	L2
Q5.	Each of the Java files on this page represents a complete source file. Your job is to play compiler and determine whether each of these files will compile. If they won't compile, how would you fix them? a. <pre> class Books { String title; String author; } class BooksTestDrive { public static void main(String [] args) { Books [] myBooks = new Books[3]; int x = 0; myBooks[0].title = "The Grape"; myBooks[1].title = "The Java"; myBooks[2].title = "The Cookbook"; myBooks[0].author = "Tony"; myBooks[1].author = "Ram"; myBooks[2].author = "Hari"; while (x < 3) { System.out.print(myBooks[x].title); System.out.print(" by "); System.out.println(myBooks[x].author); x = x + 1; } } } </pre> b. <pre> class Hobbits { String name; public static void main(String [] args) { Hobbits [] h = new Hobbits[3]; int z = 0; while (z < 4) { z = z + 1; h[z] = new Hobbits(); h[z].name = "bilbo"; if (z == 1) { </pre>	3*2	CO1	L3
			CO2	L3

Roll No. _____

Name _____

	<pre> h[z].name = "frodo"; } if (z == 2) { h[z].name = "sam"; } System.out.print(h[z].name + " is a "); System.out.println("good Hobbit name"); } } } </pre>			
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Course Outcome wise Marks Distribution



Bloom Level Wise Marks Distribution

